A place For Dad

Week 1: System Design, Core Feature Development, and Escrow Integration

Day 1: Finalize System Design & Features Breakdown:

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- **Task**:
  - Review the system goals and features (seller, buyer, admin roles).
  - Finalize the **database schema** in PostgreSQL (including tables for users, listings, transactions, and payments).
  - Define the **escrow flow** for payments (via Stripe), handling commission and fund release.
  - Create a project plan with a daily task breakdown (already started).
  - **Setup project repository** and initial structure (FastAPI for backend, PostgreSQL, MongoDB).

Day 2: Database Setup & Models

- **Task**:
  - Set up **PostgreSQL** database for:
    - **Users** (with role assignment: seller, buyer, admin)
    - **Listings** (with status management: draft, pending, approved, rejected)
    - **Transactions** (including escrow status and commission info)
  - Implement **MongoDB** for real-time chat and wishlists (store chat messages, listing saves, and buyer interactions).
  - **Create migration scripts** for tables and database schema using **Alembic**.

Day 3: User Authentication and Role Management

- **Task**:
  - Integrate **Clerk** for **authentication** and **role-based access control** (RBAC).
  - Implement **JWT authentication** for stateless login sessions (user registration, login, role assignment).
  - Develop **user registration API** and **JWT token issuance**.

Day 4: Seller Features - Listings CRUD

- **Task**:
  - Implement **Seller API** for:
    - **Create listing** (status: pending)
    - **Update own listing** (status: pending)
    - **View own listings**
    - **Delete listing** (soft delete)
  - Set up **S3 bucket integration** to store listing images and videos.
  - Implement **AWS S3 file upload** API for listing media.

Day 5: Buyer Features - Listings & Wishlist

- **Task**:
  - Implement **Buyer API** for:
    - **View approved listings** only.
    - **Search & filter listings** (by title, price, capacity, etc.).
    - **Add/remove listings to wishlist** (stored in MongoDB).
  - Develop **Buyer Dashboard** to display saved listings and search results.

Day 6: Admin Features - Listings Management

- **Task**:
    - Implement **Admin API** for:
        - **Approve/reject listings**.
        - **View and search all listings**.
        - **Soft delete listings** (globally).
    - Implement **Admin Dashboard** for listing moderation.

Day 7: Stripe Payment System Integration (Escrow Setup)

- **Task**:
    - Set up **Stripe Connect** for handling payments and escrow:
        - Create **PaymentIntent** for payment processing.
        - Use **Destination Charges** to split payments (platform commission + seller's payout).
        - Integrate **webhooks** to handle payment success and failure events.
    - Implement **Escrow flow**:
        - Funds are held temporarily in escrow until buyer confirms property (or conditions are met).
        - Commission is deducted before transferring funds to the seller's account.

Week 2: Escrow System Testing, Real-Time Features, Admin/Buyer/Seller Dashboards, and Deployment

Day 8: Real-Time Chat and Notifications (WebSockets)

- **Task**:
  - Integrate **WebSocket-based messaging** for chat between buyers and sellers.
  - Implement **real-time notifications** for chat messages, listing status changes, and admin actions.
  - Store chat messages in **MongoDB** for easy retrieval and real-time communication.

Day 9: Payment Flow Testing & Escrow Verification

- **Task**:
  - Conduct **unit tests** for the **Escrow Payment System**:
    - Verify that payments are held in escrow.
    - Ensure that commission is deducted correctly.
    - Confirm that funds are only released to the seller when conditions are met (buyer confirmation, admin approval, etc.).
  - Test **webhook handling** for Stripe events such as payment success, transfer creation, and failure events.

Day 10: Seller Dashboard Enhancements

- **Task**:
  - Add features to **Seller Dashboard**:
    - Track listing status (pending, approved, rejected).
    - View analytics (views, likes) on listings.
    - View payment status (escrow status, commission, release funds).
    - Integrate with **Stripe** to show pending payments and released funds.

Day 11: Buyer Dashboard Enhancements

- **Task**:
    - Enhance **Buyer Dashboard** to:
        - Display approved listings.
        - Show wishlist items.
        - Schedule meetings with sellers (Google Meet link integration).
        - Show payment and escrow status for ongoing transactions.

Day 12: Admin Dashboard Enhancements

- **Task**:
    - Enhance **Admin Dashboard** to:
        - View and approve/reject listings.
        - View all transactions, including escrow payments.
        - Monitor scheduled calls between buyers and sellers.
        - Track transaction history (including escrow status and payments).

Day 13: Testing User Flow (End-to-End)

- **Task**:
    - Conduct **end-to-end testing** of the user flow:
        - **Buyer** searches for listings, views approved listings, and makes payments (escrow).
        - **Seller** creates a listing, manages it, and receives payment after confirmation.
        - **Admin** moderates listings and manages users.

  - Test integration of **Stripe Connect**, **webhooks**, and **escrow payment flow**.

Day 14: Deployment & Final Testing

- **Task**:
  - Final round of **user acceptance testing (UAT)**.
  - Ensure that **Stripe keys** are set correctly for live transactions.
  - Deploy the application to production:
    - **Backend (FastAPI)**: Deploy on **AWS EC2** for easy scaling.
    - **Database (PostgreSQL & MongoDB)**: Ensure cloud databases are configured.
    - **S3**: Confirm that AWS S3 is properly storing media files.
  - Monitor the platform in production for any issues.

Milestones to Track Progress:

1. **Day 7**: Basic payment flow and escrow functionality is integrated.
2. **Day 10**: All seller and buyer dashboard features are implemented and tested.
3. **Day 12**: Admin dashboard is fully functional, including payment and escrow management.
4. **Day 14**: Platform deployed with full escrow payment flow operational.